

Physics Guess Paper Class 11 Federal Board, 2026

MCQs [Exercise MCQs + Model + Past Papers of 2025 (Mandatory)]

Chapter 1: Physics Quantities & Measurements

1.3.1 Advantages of Dimensions

1.3.2 Limitations of Dimensional Analysis

1.4 PRECISION AND ACCURACY

1.5 UNCERTAINTIES

1.2 Express the units of the following derived quantities in terms of base units. (a) Force (b) Work (c) Power (d) Pressure (e) Electric charge.

1.3 Why is it important to use an instrument with the smallest resolution?

1.4 What is the importance of increasing the number of readings in an experiment?

1.5 What is the difference between precision and accuracy?

1.6 What is the principle of homogeneity of dimensions?

1.7 A ball is thrown in the air and 5 different students are individually measuring the time it takes to fall back down using stopwatches. The times obtained by each student are the following: 6.2 s, 6.0 s, 6.4 s, 6.1 s, 5.8 s. (i) What is the uncertainty of the results? (ii) How should the resulting time be expressed?

1.8 The energy of a photon is given by $E = hf$, where f is frequency. Find the dimensions of Planck's constant h .

1.9 Justify why all measurements contain some uncertainty.

Examples of All Chapters are
Mandatory (Only Formative)

1.2 Determine the dimensions of each of the following quantities.

a) $\frac{v^2}{ax}$ b) $\frac{at^2}{2}$

(Ans: (a) No, (b) [L])

1.3 If $A = \frac{X^2}{Y^2Z}$, then find the percentage uncertainty in A. The percentage uncertainties in X, Y and Z are 1 %, 1 % and 2 % respectively.

(Ans: 6 %)

1.4 A spherical ball of radius r experiences a resistive force F due to the air as it moves through it at speed v . The resistive force F is given by the expression

$$F = c r v$$

Where c is constant. By using dimensions, derive the SI base unit of the constant c .

(Ans: $\text{kg m}^{-1}\text{s}^{-1}$)

1.5 The pressure (P) at a depth (h) in an incompressible fluid of density (ρ) is given by

$$P = \rho g h$$

Where g is acceleration due to gravity. Check the homogeneity of this equation.

1.8 The current passing through a resistor $R = (13 \pm 0.5) \Omega$ is $I = (3 \pm 0.1) \text{ A}$.

a) Calculate the power consumed (correct to one significant figure).

b) Find the percentage uncertainty of the current passing through the resistor.

c) Find the percentage uncertainty of the resistance.

d) Find the absolute uncertainty of the electrical power.

(Ans: 117 W, 3 %, 3.84 %, 11.7 W)

Chapter 2: Vectors

2.2 PRODUCT OF VECTORS

Give short answers of the following questions.

2.1 If the cross product of two vectors vanishes, what will you say about their orientation?

- 2.2 Find the dot product of unit vectors with each other at (a) 0° and (b) 90° .
- 2.3 Show that scalar product obeys commutative property.
- 2.4 Solve by using the properties of dot and cross product: (a) $\hat{i} \cdot (\hat{j} \times \hat{k})$ (b) $\hat{j} \times (\hat{j} \times \hat{k})$?
- 2.5 If both the dot product and the cross product of two vectors are zero. What would you conclude about the individual vectors?
- 2.6 What are rectangular components of a vector? How they can be found?
- 2.7 Give two examples for each of the scalar and vector product.
- 2.8 Show that: $\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$.
- 2.9 What units are associated with the unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$?
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- 2.1 If the magnitude of cross product between two vectors is $\sqrt{3}$ times the dot product, find angle between them. (Ans: 60°)
- 2.2 A force is acting on a body making an angle of 30° with the horizontal. The horizontal component of the force is 20 N. Find the force. (Ans: 23.1 N)
- 2.3 A vector $\mathbf{F}$ having magnitude 5.5 N makes 10° with x-axis and a vector $\mathbf{r}$ with magnitude 4.3 m makes 80° with x-axis. What is the magnitude of their dot and cross products? (Ans: 8.1 N m and 22.2 N m)
- 2.4 The magnitude of dot and cross product of two vectors are $6\sqrt{3}$ and 6, respectively. Find the angle between the vectors. (Ans: $\theta = 30^\circ$)
- 2.5 A force of 200 N is making an angle of 30° with the x-axis. Find the horizontal and vertical components of this force. (Ans: 173.2 N and 100 N)
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Chapter 3: Translatory Motion

3.2 PROJECTILE MOTION

3.2 You throw a small ball vertically up in the air. How are the velocity and acceleration of the ball oriented with respect to one another (a) when the ball is moving upward (b) when the ball is moving downward?

3.3 For a projectile motion, is the velocity ever zero? Is the acceleration ever zero?

3.4 Draw a diagram showing the velocity and acceleration of a projectile at several points along its path, assuming (a) the projectile is launched horizontally and (b) the projectile is launched at an angle θ with the horizontal.

3.5 An aeroplane while flying horizontally drops a bomb when it reaches exactly above the target, but misses it. Explain why?

3.6 How does air resistance affect projectile motion?

3.2 Find the angle of projection of a projectile for which the maximum height and corresponding range are equal. (Ans: 76°)

3.4 A projectile is launched with an initial speed of 21.8 m s^{-1} at an angle of 35° above the horizontal. Determine:

(a) the time of flight of the projectile.

(b) the peak height of the projectile.

(c) the horizontal displacement of the projectile. (Ans: 2.55 s, 7.98 m, 45.6 m)

3.5 A projectile is launched horizontally from the top of a 45.2 m high cliff and lands a distance of 17.6 m from the base of the cliff. Determine the magnitude of the launch velocity.

(Ans: 5.79 m s^{-1})

Chapter 4: Rotational and Circular Motion

4.2 CENTRIPETAL ACCELERATION AND CENTRIPETAL FORCE

4.4 ANGULAR MOMENTUM

4.5 TORQUE AND ANGULAR ACCELERATION

4.7 ARTIFICIAL GRAVITY

All Short Questions from the exercise

4.1 What will be the angular velocity of fly wheel of an engine if it completes 3000 revolutions in a minute? (Ans. 314 rad s^{-1})

4.3 A PT teacher rotates his stick at the axis that passes through its centre. If mass of the stick is 200 g and its length is 0.8 m, find its moment of inertia. (Ans. 0.01 kg m^2)

4.4 A football of mass 450 g rotates with an angular speed of 10 rev s^{-1} . If its radius is 11 cm, compute its angular momentum. (Ans. 0.137 J s)

4.7 A wheel-shape space station provides an artificial gravity of 5.00 m s^{-2} to its inhabitants. If it has a diameter of 100 m, find its angular speed in rpm. (Ans. 3 rpm)

4.8 The minute hand of a watch is 2 cm long. If it travels 9.4 cm of length, how many radians will it travel? (Ans. 4.7 rad)

4.9 How many revolutions does the fidget spinner make after being flicked with an initial angular velocity of 10 revolutions per second, and coming to rest in 5 seconds? (ignore air resistance) (Ans. 25 revolutions)

Chapter 5: Work and Kinetic Energy

5.2 CONSERVATIVE & NON-CONSERVATIVE FIELDS

5.4.1 Work - Energy Theorem in Resistive Medium

All Short Questions from the exercise

5.3 A ball of mass 100 g is released from a height of 30 m. If the ball encounters an air resistance of 0.4 N, find the kinetic energy of the ball just before striking the ground.

5.6 A car engine operates with 25 % efficiency and produces 50 kJ of useful work. How much fuel energy did the car consume? (Ans: 200 kJ)

5.7 A refrigerator uses 600 W of electrical power and has an efficiency of 15 %. How much cooling power (useful energy) does the refrigerator produce in one hour? (Ans: 324 kJ)

Chapter 6: Fluid Mechanics

6.1 UPTHRUST AND ARCHIMEDES PRINCIPLE

6.5 BERNOULLI'S EQUATION

Learn Conceptual Questions from this chapter

All Short Questions from the exercise

6.1 The settling speed of a small fog droplet in air is found to be 2.98 mm s^{-1} . Find out the radius of the droplet. The coefficient of viscosity of air is $1.9 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$. (Ans: $5.1 \times 10^{-6} \text{ m}$)

6.3 Blood is pumped from the heart at a rate of 5.0 L min^{-1} into the aorta (of radius 1.0 cm). Determine the speed of blood through the aorta. (Ans: 27 cm s^{-1})

6.4 Water flows through a pipe with an internal diameter 4 cm at a speed of 2 m s^{-1} . What should be the diameter of the nozzle, if the water is to emerge out with a speed of 4 m s^{-1} . (Ans: 2.8 cm)

6.5 A hose lying on the ground has water coming out of it at a speed of 5.4 m s^{-1} . You lift the nozzle of the hose to a height of 1.3 m above the ground. At what speed does the water now come out of the hose? (Ans: 1.9 m s^{-1})

6.8 A hole is drilled at the bottom of a full bucket of water of height h . What will be the velocity of water exiting from the hole? (Ans: $\sqrt{2gh}$)

6.9 The efflux speed of oil from a narrow hole of a tank is 33.8 m s^{-1} , at pressure of $4.12 \times 10^5 \text{ Pa}$. Calculate the density of the oil? (Ans: 0.72 kg m^{-3})

Chapter 7: Deformation of Solids

7.2.3 Stress-Strain Curve

7.3 ELASTIC POTENTIAL ENERGY

All Short Questions from the exercise

7.4 A rod with a cross-sectional area of 90 mm^2 and a length of 3 m . If a stress of 300 MPa is applied to stretch the rod, then find the strain energy if Young's modulus of the rod is 200 GPa . (Ans: 12.15 J)

7.5 A force of 500 N is applied to one end of a cylindrical steel rod of diameter 50 cm . What is the tensile stress? (Ans: $2.5 \times 10^5 \text{ N m}^{-2}$)

Chapter 8: Heat and Thermodynamics

8.2 KINETIC THEORY OF GASES

8.3.3 Gay Lussac's Law

8.4 FIRST LAW OF THERMODYNAMICS

8.8 CARNOT'S ENGINE

All Short Questions from the exercise

8.1 Calculate number of moles of air in an inflated balloon of radius 10 cm and pressure 180 kPa at room temperature. (Ans: 0.3 moles)

8.2 A heat engine operating between the freezing and boiling point of water. Find its efficiency. (Ans: 27%)

8.4 The high temperature of Carnot engine is 600 K. If engine absorbs 600 J of heat at low temperature of 400 K. Find the work done by the engine. (Ans: 200 J)

8.5 A 220 W electric immersion heater is used to boil 136 g of water. Calculate the time required to bring this water from 23.5 °C to its boiling point, ignoring any heat loss.

8.6 If a heat engine at room temperature has an efficiency 50 %, you change the heat source to increase the efficiency up to, 70 %. Find the temperature difference between the two heat sources. (Ans: 397 K)

8.7 A Carnot engine absorbs an amount Q of heat at temperature T . Find the amount of heat rejected at temperature $T/3$. (Ans: $Q/3$)

8.8 Energy 500 J is required to melt 2 g of ice at 0 °C. Find the change in entropy of 70 g water at 0 °C, if it changes into ice in a refrigerator. (Ans: -64 J K^{-1})

8.9 A gas expands from volume 1m^3 to 2m^3 at constant atmospheric pressure.
(a) Calculate the work done by the gas.

Chapter 9: Waves

9.2 DOPPLER'S EFFECT

9.6.1 Stationary Waves in a Stretched String

9.7 STATIONARY WAVES IN AIR COLUMN

Malus's law

All Shot and Numerical

Chapter 10: Electrostatics

10.4 POTENTIAL GRADIENT.

All Short Questions

Numerical : 2 to 7

Chapter 11: Electricity

11.3 ELECTROMOTIVE FORCE AND MAXIMUM POWER OUTPUT

11.7 KIRCHHOFF'S LAWS

Deriving the Equation for Resistors in Series

Deriving the Equation for Resistors in Parallel

11.8 NULL METHOD OF MEASUREMENT

All Short Questions

All Numerical

Chapter 12: Magnetism

12.4 MAGNETIC FORCE ACTING ON MOVING CHARGE IN UNIFORM MAGNETIC FIELD

12.5 MOTION OF A CHARGED PARTICLE IN A UNIFORM MAGNETIC FIELD

12.12 FARADAY'S LAW

All Short Questions

Numerical: 2,3,4 , 7,8,9

Chapter 13: Relativity

13.3 CONSEQUENCES OF THE SPECIAL THEORY OF RELATIVITY

All Short Questions & Numerical

Chapter 14: Particle Physics

14.3 MATTER AND ANTI-MATTER

14.4 QUARKS

14.5 LEPTONS

14.7 PARTICLE ANNIHILATION

All Short Questions

14.2 A carbon-14 nucleus undergoes beta-minus (β^-) decay, emitting a beta particle (electron) with a negligible mass. Calculate the energy released in this beta-minus decay process. The speed of light (c) is approximately 3.00×10^8 meters per second. (Ans: 0.17MeV)

14.3 A nucleus undergoes a gamma decay, emitting a gamma ray photon with energy of 2.00 MeV. Calculate the frequency and wavelength of this gamma ray.

(Ans: 3.02×10^{20} Hz, 9.94×10^{-13} m)